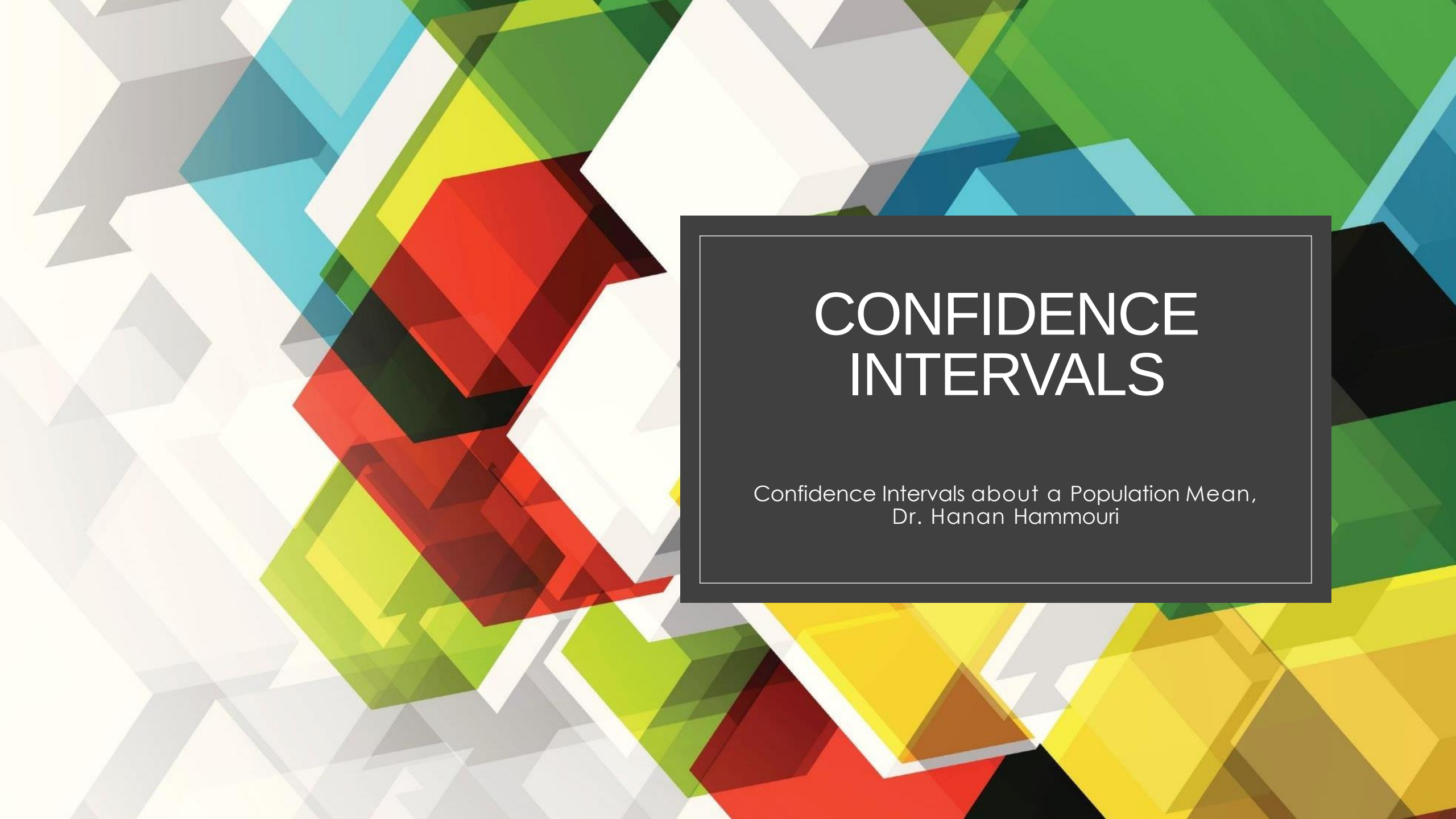




CONFIDENCE INTERVALS

Confidence Intervals about a Population Mean,
Dr. Hanan Hammouri

Point estimate

- A **point estimate** of a parameter is the value of a statistic that estimates the value of the parameter.

The sample mean, $\bar{x}$, is the **best point estimate** of the population mean, μ .

Confidence interval estimate

A **confidence interval estimate** of a parameter consists of an interval of numbers along with a probability that the interval contains the unknown parameter.

The **level of confidence** in a confidence interval is a probability that represents the percentage of intervals that will contain if a large number of repeated samples are obtained.

For example, a 95% level of confidence would mean that if 100 confidence intervals were constructed, each based on a different sample from the same population, we would expect 95 of the intervals to cover the population mean.

The construction of a confidence interval needs

- ❑ The point estimate of the population
- ❑ The level of confidence $(1-\alpha)100\%$
- ❑ Use t or z depending on the sampling distribution
- ❑ The standard deviation of the sample mean $\sigma_{\bar{x}}$ depending on the sampling distribution

What is α (alpha)?

To answer this question we are interested in (true) depending on the sample , while in reality its false(false positive).

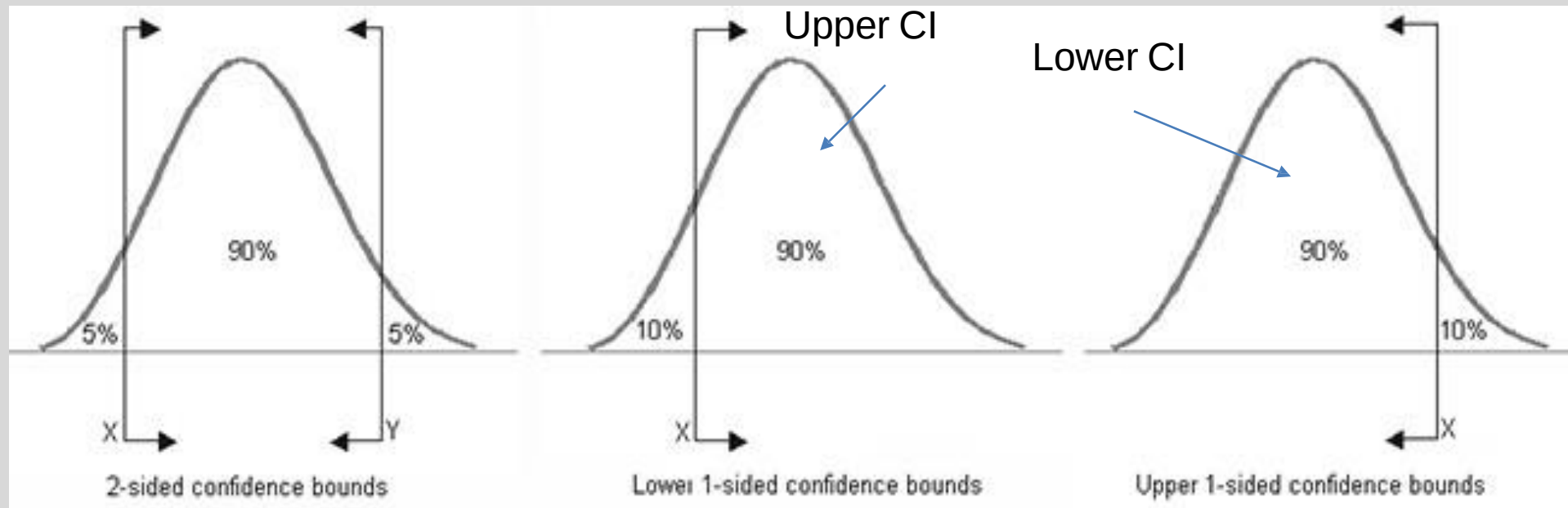
		Reality	
		True	False
Measured from the sample	True	Correct 😊	Type I False Positive
	False	Type II False Negative	Correct 😊

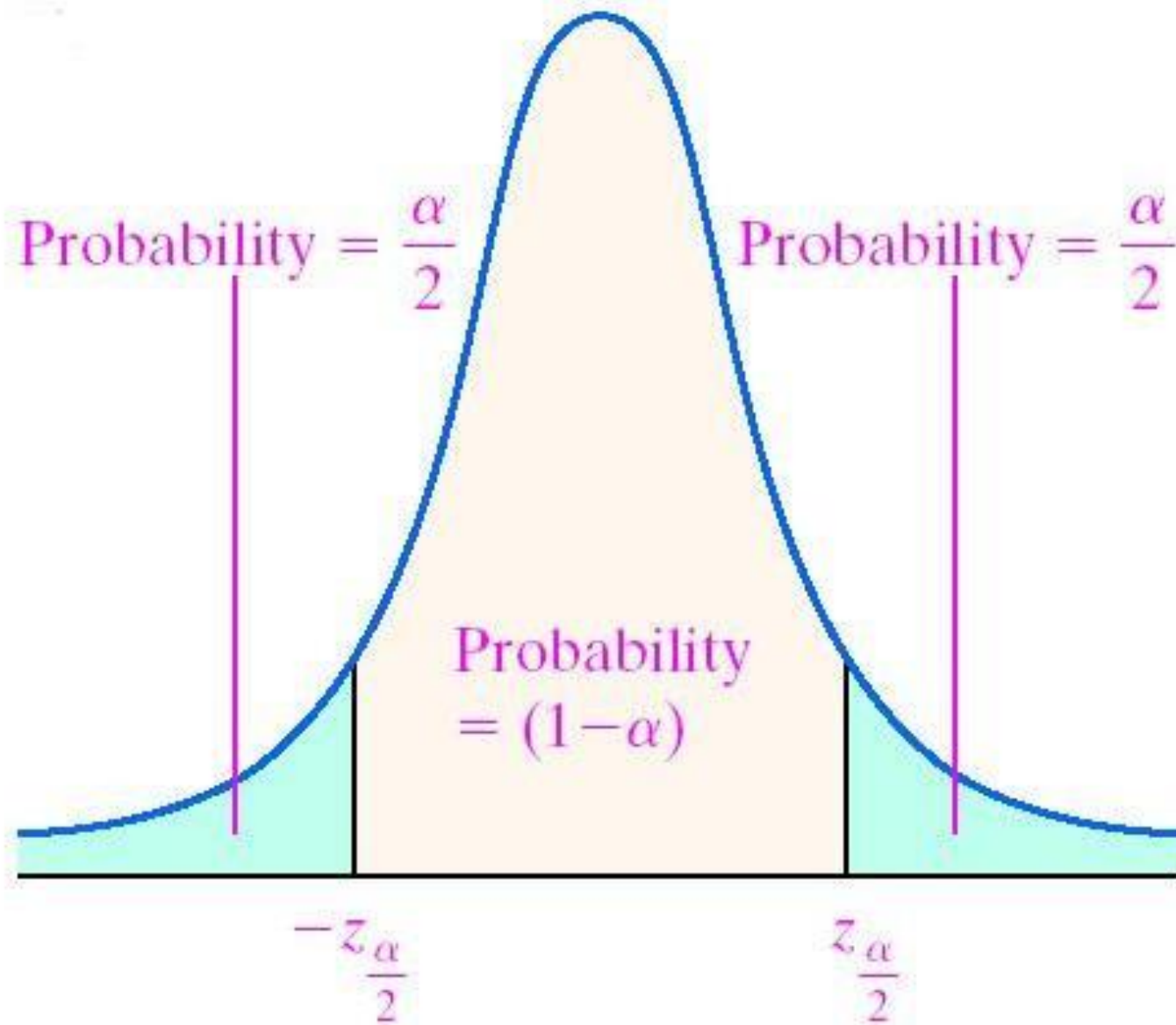
Confidence level

α alpha	Confidence level (1- α)100%	Two tailed $z_{\alpha/2}$	One tailed z_{α}
0.01	99%	2.58	2.33
0.05	95%	1.96	1.64
0.1	90%	1.64	1.28

We need to find the z-score for each confidence level. First, we need to know if the CI one sided or two sided

One sided vs two sided





Two sided confidence interval

$$\bar{x} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$P\left(\bar{x} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = (1-\alpha)100\%$$

Interpretation for confidence interval

We are $(1-\alpha)100\%$ that the confidence interval covers the true value for the mean μ

Margin of error (ME)

- Margin of error = Critical value x Standard deviation of the statistic
- In two sided case $ME = Z_{\frac{\alpha}{2}}(t_{\frac{\alpha}{2}}) \times \frac{\sigma}{\sqrt{n}}$
- In one sided case $ME = Z_{\alpha}(t_{\alpha}) \times \frac{\sigma}{\sqrt{n}}$

Example: when the sampling distribution is normal

- A physical therapist wished to estimate, with 99 percent confidence, the mean maximal strength of a particular muscle in a certain group of individuals. He is willing to assume that strength scores are approximately normally distributed with a variance of 144. A sample of 15 subjects who participated in the experiment yielded a mean of 84.3(two sided).

$$84.3 \pm 2.58(12/\sqrt{15})$$
$$(76.3, 92.3)$$

Example: when the sampling distribution is t distribution

- A physical therapist wished to estimate, with 99 percent confidence, the mean maximal strength of a particular muscle in a certain group of individuals. He is willing to assume that strength scores are approximately normally distributed. A sample of 15 subjects who participated in the experiment yielded a mean of 84.3 and a variance of 144(two sided).

$$84.3 \pm 2.977(12/\sqrt{15})$$
$$(75.08, 93.52)$$

One sided confidence intervals

A $100(1 - \alpha)\%$ **upper-confidence** bound for μ is

$$\mu \leq \bar{x} + z_{\alpha}\sigma/\sqrt{n}$$

and a $100(1 - \alpha)\%$ **lower-confidence** bound for μ is

$$\bar{x} - z_{\alpha}\sigma/\sqrt{n} \leq \mu$$

Example

- A physical therapist wished to estimate, with 99 percent confidence, the mean maximal strength of a particular muscle in a certain group of individuals. He is willing to assume that strength scores are approximately normally distributed with a variance of 144. A sample of 15 subjects who participated in the experiment yielded a mean of 84.3(one sided(lower CI with upper bound)).

$$\mu \leq 84.3 + 2.33(12/\sqrt{15})$$

$$\mu \leq 91.5192$$

$$(-\infty, 91.52)$$

Example

- A physical therapist wished to estimate, with 99 percent confidence, the mean maximal strength of a particular muscle in a certain group of individuals. He is willing to assume that strength scores are approximately normally distributed with a variance of 144. A sample of 15 subjects who participated in the experiment yielded a mean of 84.3(one sided(upper CI with lower bound)).

$$84.3 - 2.33(12/\sqrt{15}) \leq \mu$$

$$77.0808 \leq \mu$$

$$(77.08, \infty)$$

Confidence Interval for μ

Example two sided

A random sample of $n = 25$ has $\bar{X} = 50$ and $S = 8$. Form a 95% confidence interval for μ

– d.f. = $n - 1 = 24$, so

The confidence interval is

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} = 50 \pm (2.064) \frac{8}{\sqrt{25}}$$

(46.698 , 53.302)

d.f.	$\alpha = .05$ two sided
24	2.064

Confidence Interval for μ

- Example one sided

◦

- A random sample of $n = 25$ has $\bar{X} = 50$ and $S = 8$. Form a 95% upper

- confidence interval for μ

- d.f. = $n - 1 = 24$, so

- The confidence interval is

$$\bar{X} - t_{\alpha, n-1} \frac{S}{\sqrt{n}} = 50 - (1.711) \frac{8}{\sqrt{25}}$$

$$(47.26, \infty)$$

d.f.	$\alpha = .05$ one sided
24	1.711